

AI Interaction Log

CIE-1 — Computer Organisation & Operating Systems | Hardik Bhalekar

1. Planning

Prompt: Help me build the CIE-1 System Detective assignment using my own WSL machine. I need the five Linux command outputs, locality experiment, dashboard and AI log.

AI answer (summary): Use lscpu, free -h, top/htop, ps -eLf and strace -c; run row-major versus column-major; then build a dashboard from measured data.

2. Locality experiment

Prompt: Help me implement and explain a row-major versus column-major 2D-array experiment.

AI answer (summary): C stores 2D arrays in row-major order. Row traversal accesses adjacent addresses and generally benefits from spatial locality; column traversal uses a larger stride.

Machine evidence: three runs were 0.091525/0.509816 s, 0.290383/1.848290 s and 0.093486/0.504370 s. Mean = 0.158465 s versus 0.954159 s; ratio = 6.02x.

3. Proven AI error

Prompt: Generate/fix the Streamlit dashboard for the System Detective project and include the required command outputs and locality experiment.

AI answer: AI-generated/revised app.py code was supplied.

Actual execution error: Python reported an unterminated triple-quoted string at line 1383, detected through line 1391. The generated source therefore did not run.

Correction: I inspected the reported line range, fixed the unmatched multiline string and reran the dashboard. This is the required machine-verified AI error.

4. Live data verification

Prompt: Make the dashboard show my actual CPU, memory, processes and system-call information.

AI answer (summary): Read live Linux commands/sources and do not hardcode sample values.

Evidence: the captures show x86_64, Intel Core i3-4010U, 4 logical CPUs, 6.1 GiB memory and 2.0 GiB swap.

Reflection

AI was useful for implementation and explanation, but the generated code was tested on the actual machine. The syntax error was caught by execution, and the locality result was measured directly rather than invented or copied from the sample.